

Advanced OOP Practical Java Programs

Single Inheritance

```
class Animal {
    void eat() {
        System.out.println("Animal is eating");
    }
}

class Dog extends Animal {
    void bark() {
        System.out.println("Dog is barking");
    }
}

public class SingleInheritanceDemo {
    public static void main(String[] args) {
        Dog d = new Dog();
        d.eat();
        d.bark();
    }
}
```

Multilevel Inheritance

```
class Person {
    void display() {
        System.out.println("Person Class");
    }
}

class Student extends Person {
    void showStudent() {
        System.out.println("Student Class");
    }
}

class CollegeStudent extends Student {
    void showCollege() {
        System.out.println("College Student");
    }
}

public class MultilevelDemo {
    public static void main(String[] args) {
        CollegeStudent c = new CollegeStudent();

        c.display();
        c.showStudent();
        c.showCollege();
    }
}
```

Method Overloading

```
class Overload {

    int add(int a, int b) {
        return a + b;
    }

    double add(double a, double b) {
        return a + b;
    }

    public static void main(String[] args) {
        Overload obj = new Overload();

        System.out.println(obj.add(10, 20));
    }
}
```

```

        System.out.println(obj.add(2.5, 3.5));
    }
}

```

Constructor Program

```

class Student {
    int roll;
    String name;

    Student(int r, String n) {
        roll = r;
        name = n;
    }

    void display() {
        System.out.println(roll + " " + name);
    }
}

public class ConstructorDemo {
    public static void main(String[] args) {
        Student s1 = new Student(1, "Om");
        Student s2 = new Student(2, "Rahul");

        s1.display();
        s2.display();
    }
}

```

Interface Program

```

interface Shape {
    void draw();
}

class Circle implements Shape {
    public void draw() {
        System.out.println("Drawing Circle");
    }
}

public class InterfaceDemo {
    public static void main(String[] args) {
        Circle c = new Circle();
        c.draw();
    }
}

```

Array Program

```

public class ArrayDemo {
    public static void main(String[] args) {
        int arr[] = {10, 20, 30, 40, 50};

        int sum = 0;

        for(int i = 0; i < arr.length; i++) {
            System.out.print(arr[i] + " ");
            sum += arr[i];
        }

        System.out.println();
        System.out.println("Sum = " + sum);
    }
}

```

Exception Handling

```
public class ExceptionDemo {  
    public static void main(String[] args) {  
        try {  
            int a = 10;  
            int b = 0;  
  
            int result = a / b;  
  
            System.out.println(result);  
        } catch(Exception e) {  
            System.out.println("Exception Handled");  
        }  
    }  
}
```

Calculator Program

```
import java.util.Scanner;  
  
public class Calculator {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int a, b;  
  
        System.out.print("Enter first number: ");  
        a = sc.nextInt();  
  
        System.out.print("Enter second number: ");  
        b = sc.nextInt();  
  
        System.out.println("Addition = " + (a + b));  
        System.out.println("Subtraction = " + (a - b));  
        System.out.println("Multiplication = " + (a * b));  
  
        try {  
            System.out.println("Division = " + (a / b));  
        } catch(Exception e) {  
            System.out.println("Cannot divide by zero");  
        }  
  
        sc.close();  
    }  
}
```

Thread Program

```
class MyThread extends Thread {  
    public void run() {  
        for(int i = 1; i <= 5; i++) {  
            System.out.println("Thread Running " + i);  
        }  
    }  
}  
  
public class ThreadDemo {  
    public static void main(String[] args) {  
        MyThread t = new MyThread();  
  
        t.start();  
    }  
}
```

```
}
```

Bank Management System

```
class Bank {  
    int balance = 5000;  
  
    void deposit(int amount) {  
        balance += amount;  
        System.out.println("Balance after deposit: " + balance);  
    }  
  
    void withdraw(int amount) {  
        balance -= amount;  
        System.out.println("Balance after withdraw: " + balance);  
    }  
}  
  
public class BankDemo {  
    public static void main(String[] args) {  
        Bank b = new Bank();  
  
        b.deposit(1000);  
        b.withdraw(500);  
    }  
}
```

Student Result Program

```
class Student {  
    int marks1 = 80;  
    int marks2 = 75;  
    int marks3 = 90;  
  
    void result() {  
        int total = marks1 + marks2 + marks3;  
  
        double avg = total / 3.0;  
  
        System.out.println("Total = " + total);  
        System.out.println("Average = " + avg);  
    }  
}  
  
public class StudentResult {  
    public static void main(String[] args) {  
        Student s = new Student();  
  
        s.result();  
    }  
}
```